AOFC Exercise 2 — 28.10.2025

Ex. 4 a) energy velocity $v_e = \frac{S}{u}$ ← S: poyuting-vector; u: density of energy

$S = \vec{E} \times \vec{H} \quad \left[\frac{W}{m^2}\right]$

$u = 1/2(\varepsilon_0 \varepsilon_r E^2 + \mu_0 H^2)$

Energy vel.: $\frac{\left[\frac{W}{m^2}\right]}{\left[\frac{Ws}{m^3}\right]} = \left[\frac{m}{s}\right]$

x-polarized wave: $\vec{E} = \vec{e}_x E_0 e^{i(\omega t - \vec{k}\vec{r})} \quad \vec{H} = \vec{e}_y H_0 e^{i(\omega t - \vec{k}\vec{r})}$

$\vec{k} = \begin{pmatrix} 0 \\ 0 \\ k_z \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 0 \\ \beta(\omega) \end{pmatrix}$ ← dispersive material

β: propagation constant

$N = n(\lambda) - \lambda \cdot \frac{dn(\lambda)}{d\lambda} \neq n \qquad v_{gr} \neq v_{ph} = \frac{c_0}{n}$

$v_{gr} = \frac{1}{\frac{\partial \beta(\omega)}{\partial \omega}} \qquad \beta(\omega) = \frac{\omega \cdot n(\omega)}{c_0}$

$v_e = \frac{E_0 H_0}{1/2 (\varepsilon_r \varepsilon_0 E_0^2 + \mu_0 H_0^2)} \qquad v_e < v_{gr}$

$\nabla \times \vec{E} = i\omega\mu_0 \vec{H} = -i\vec{k} \times \vec{E}$ 3rd Maxwell

$\nabla \times \vec{H} = i\omega\varepsilon_0\varepsilon_r \vec{E} = -i\vec{k} \times \vec{H}$ 4rd Maxwell

plane wave relationship

$$\begin{aligned} \omega\mu_0 \vec{H} &= \vec{k} \times \vec{E} \\ -\omega\varepsilon_0\varepsilon_r \vec{E} &= \vec{k} \times \vec{H} \end{aligned}$$

$\partial\omega$ $\partial\vec{k}$, $\partial\vec{E}$, $\partial\vec{H}$ are changing!

$\partial\vec{k} \times \vec{E} + \vec{k} \times \partial\vec{E} = \partial\omega\mu_0\vec{H} + \omega\mu_0 \partial\vec{H} \quad | \cdot E$

$\partial\vec{k} \times \vec{H} + \vec{k} \times \partial\vec{H} = -\partial\omega\varepsilon_0\varepsilon_r \vec{E} - \omega\varepsilon_0\varepsilon_r \quad | \cdot H$

$\partial\vec{k} \cdot (\vec{E} \times \vec{H}) + \vec{k} \cdot (\partial\vec{E} \times \vec{H}) = \partial\omega\mu_0 \vec{H} \cdot \vec{H} + \omega\mu_0 \partial\vec{H} \cdot \vec{H}$ (1)

$-\partial\vec{k} \cdot (\vec{E} \times \vec{H}) + \vec{k} \cdot (\partial\vec{H} \times \vec{E}) = -\partial\omega\varepsilon_0\varepsilon_r \vec{E}\vec{E} - \omega\varepsilon_0\varepsilon_r \partial\vec{E}\vec{E}$ (2)

(1)-(2): $2\partial\vec{k} \cdot (\vec{E} \times \vec{H}) = 2\omega(\mu_0 \vec{H} \cdot \vec{H} + \varepsilon_0\varepsilon_r \vec{E}\vec{E})$

$2\partial\beta E_0 H_0 = 2\omega(\mu_0 H_0^2 + \varepsilon_0\varepsilon_r E^2) \Rightarrow \frac{1}{\frac{\partial\beta}{\partial\omega}} = v_{gr} = \frac{E_0 H_0}{1/2(\varepsilon_0\varepsilon_r E_0^2 + \mu_0 H^2)}$

b) Sellmeier equation: $n^2 = 1 + \frac{B_1}{\omega_1^2 - \omega^2} + \frac{B_2}{\omega_2^2 - \omega^2}$

$\omega_2 \ll \omega$ $\quad n^2 = 1 + \frac{B_1}{\omega_1^2 - \omega^2} - \frac{B_2}{\omega^2}$

$B_1 = \omega_e \cdot \omega_d \quad \omega_1 = \omega_e, \quad B_2 = \omega_L^2$

$$n = \sqrt{\left(1 + \frac{\omega_e \omega_d}{\omega_e^2 - \omega^2} - \frac{\omega_L^2}{\omega^2}\right)} = \sqrt{\left(1 + \frac{f_e f_d}{f_e^2 - f^2} - \frac{f_L^2}{f^2}\right)} = n(f)$$

$f_e = 32.12 \cdot 10^{14}\,Hz, \quad f_d = 35.31 \cdot 10^{14}\,Hz, \quad f_L = 0.31 \cdot 10^{14}\,Hz$

$\lambda = 1.55\,\mu m \qquad f = \frac{c_0}{\lambda} = \frac{3 \cdot 10^8\,m/s}{1.55\,\mu m} = 193.55\,THz$

$$n = \sqrt{\left(1 + \frac{32.12 \cdot 10^{14}\,Hz \cdot 35.31 \cdot 10^{14}\,Hz}{(32.12 \cdot 10^{14}\,Hz)^2 - (193.55 \cdot 10^{12}\,Hz)^2} - \frac{(0.31 \cdot 10^{14}\,Hz)^2}{(193.55 \cdot 10^{12}\,Hz)^2}\right)} = 1.441$$

$$n(\lambda) = \sqrt{\left(1 + \frac{\lambda_e/\lambda_d \cdot \lambda^2}{\lambda^2 - \lambda_e^2} - \frac{\lambda^2}{\lambda_L^2}\right)}$$

c) Group index $N = n_g = n(\lambda) - \lambda \frac{dn(\lambda)}{d\lambda}$

$$\frac{dn}{d\lambda} = \frac{-\frac{2\lambda_e/\lambda_d \cdot \lambda_e^2 \cdot \lambda}{(\lambda^2 - \lambda_e^2)^2} - \frac{2\lambda}{\lambda_L^2}}{2\sqrt{\left(1 + \frac{\lambda_e/\lambda_d \cdot \lambda}{\lambda^2 - \lambda_0^2} - \frac{\lambda^2}{\lambda_L^2}\right)}} = \frac{1}{n}\left[\frac{-\lambda_e^3/\lambda_d \cdot \lambda}{(\lambda^2 - \lambda_e^2)^2} - \frac{\lambda}{\lambda_L^2}\right]$$

$$= \frac{1}{1.441}\left[\frac{\frac{(0.0934\,\mu m)^3}{0.085\,\mu m} \cdot 1.55\,\mu m}{\{(1.55\,\mu m)^2 - (0.0934\,\mu m)^2\}^2} - \frac{1.55\,\mu m}{(9.6774\,\mu m)^2}\right] = -0.01328\ 1/\mu m$$

$$N = 1.441 - 1.55\,\mu m\,(-0.01328\ 1/\mu m) = 1.462$$

$$v_{gr} = \frac{1}{\frac{2\beta}{2\omega}} \quad \text{with } \beta(\omega) = \frac{\omega \cdot n(\omega)}{c_0} \Rightarrow \frac{2\beta}{2\omega} = \frac{1}{c_0}\left[n(\omega) + \omega \frac{dn(\omega)}{d\omega}\right]$$

$$= \frac{1}{c_0}\underbrace{\left[n(f) + f \frac{dn(f)}{df}\right]}_{N}$$

$$v_{gr} = \frac{1}{\frac{N}{c_0}} = \frac{3 \cdot 10^8 \frac{m}{s}}{1.462} = 2.05 \cdot 10^8 \frac{m}{s}$$

$$v_{ph} = \frac{c_0}{n} = 2.08 \cdot 10^8 \frac{m}{s}$$

Dispersion coefficient

GVD → group velocity dispersion

$$\beta(\omega)\Big|_{\omega=\omega_0} = \beta(\omega_0) + \beta_1(\omega - \omega_0) + 1/2\,\beta_2(\omega - \omega_0)^2 + \ldots$$

$\beta = \beta(\omega_0)$

$\beta_1 = \frac{2\beta}{2\omega} = 1/v_g$

$\beta_2 = \frac{2^2\beta}{2\omega^2} = GVD$

in fiber optics mostly D is used instead of β_2

$$D = -\left(2\pi \frac{c}{\lambda^2}\right) \cdot \beta_2 = -\left(\frac{\lambda}{c}\right) \cdot \frac{d^2 n}{d\lambda^2}$$

Other way of denoting D

Group delay $\tau = \frac{1}{v_{gr}} \Rightarrow D = \frac{2\tau}{2\lambda} = \frac{1}{c_0}\frac{dN}{d\lambda} = -\frac{\lambda}{c_0}\frac{d^2 n(\lambda)}{d\lambda^2}$ $\left[\frac{ps}{nm \cdot km}\right]$